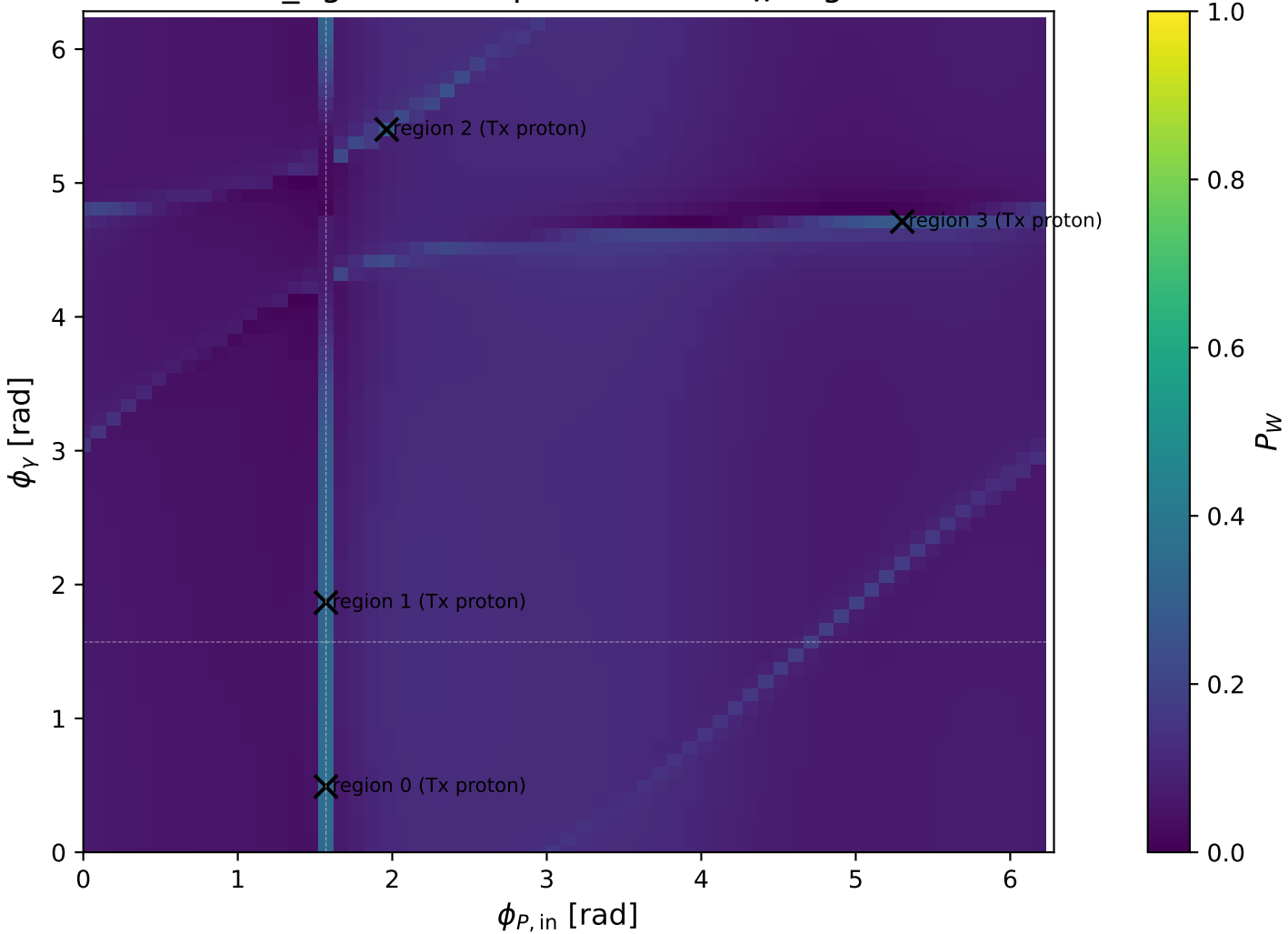


low_Egamma Tx proton: max P_W regions

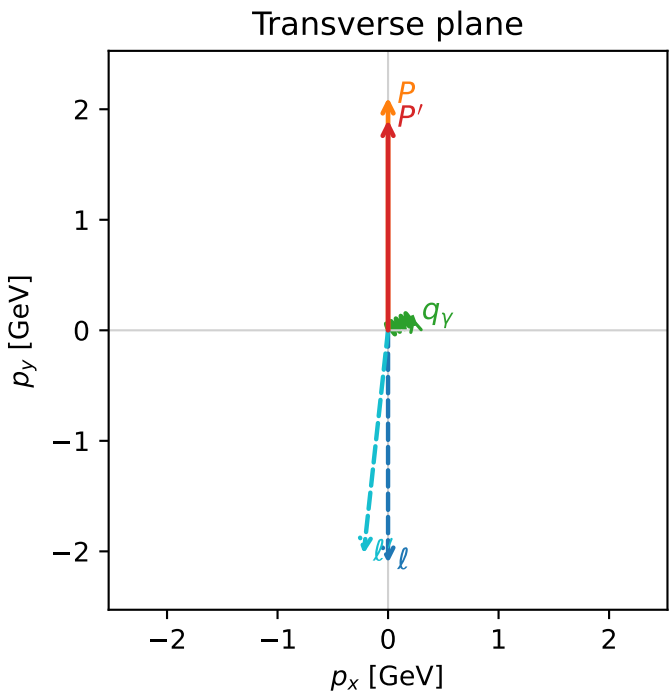
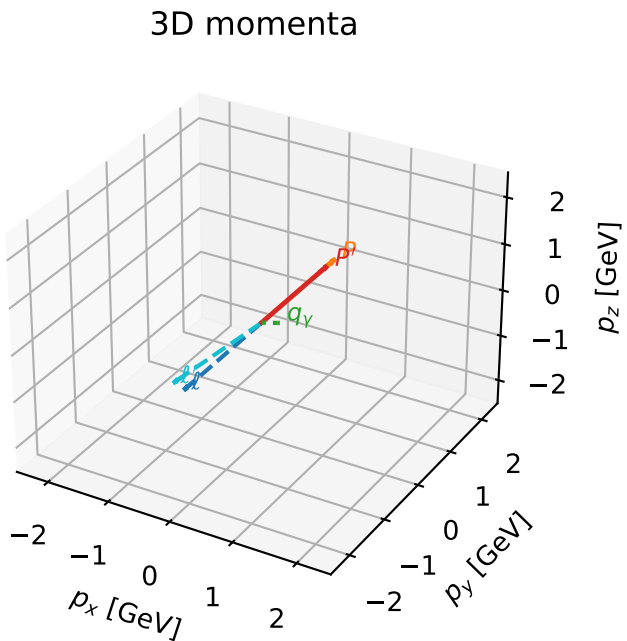


Tx proton characteristicmax P_W configuration

w_purity_low_egamma_tx_proton_region_0 P_W=0.353641
region: low_Egamma_Tx_proton_region_0
outgoing-state purity: 0.503326
kinematic point: high_s_high_theta_in_low_Egamma
incoming state: $\frac{1}{2}\sum_{h_l}\rho(h_l, T_{x,p})$

$s=21.5158$, $\sqrt{s}=4.63852$, $|\vec{P}|=2.10694$, $|\vec{P}'|=1.90379$
 $\theta_{in}=1.57079$, $\phi_l=4.71239$, $\phi_P=1.5708$
 $E_\gamma=0.25$, $\phi_\gamma=0.490874$
 $Q^2=0.0515301$, $x_B=0.0776088$, $t=-0.00742145$, $W^2=1.49229$, $y=0.033814$
 $\theta(l', \gamma)=124.349$ deg

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 2.3322$ $p_x= -0.0000$ $p_y= -2.1069$ $p_z= -0.0000$
 P $E= 2.3063$ $p_x= 0.0000$ $p_y= 2.1069$ $p_z= 0.0000$
 l' $E= 2.2662$ $p_x= -0.2205$ $p_y= -2.0216$ $p_z= 0.0000$
 P' $E= 2.1223$ $p_x= 0.0000$ $p_y= 1.9038$ $p_z= 0.0000$
 q_γ $E= 0.2500$ $p_x= 0.2205$ $p_y= 0.1178$ $p_z= 0.0000$



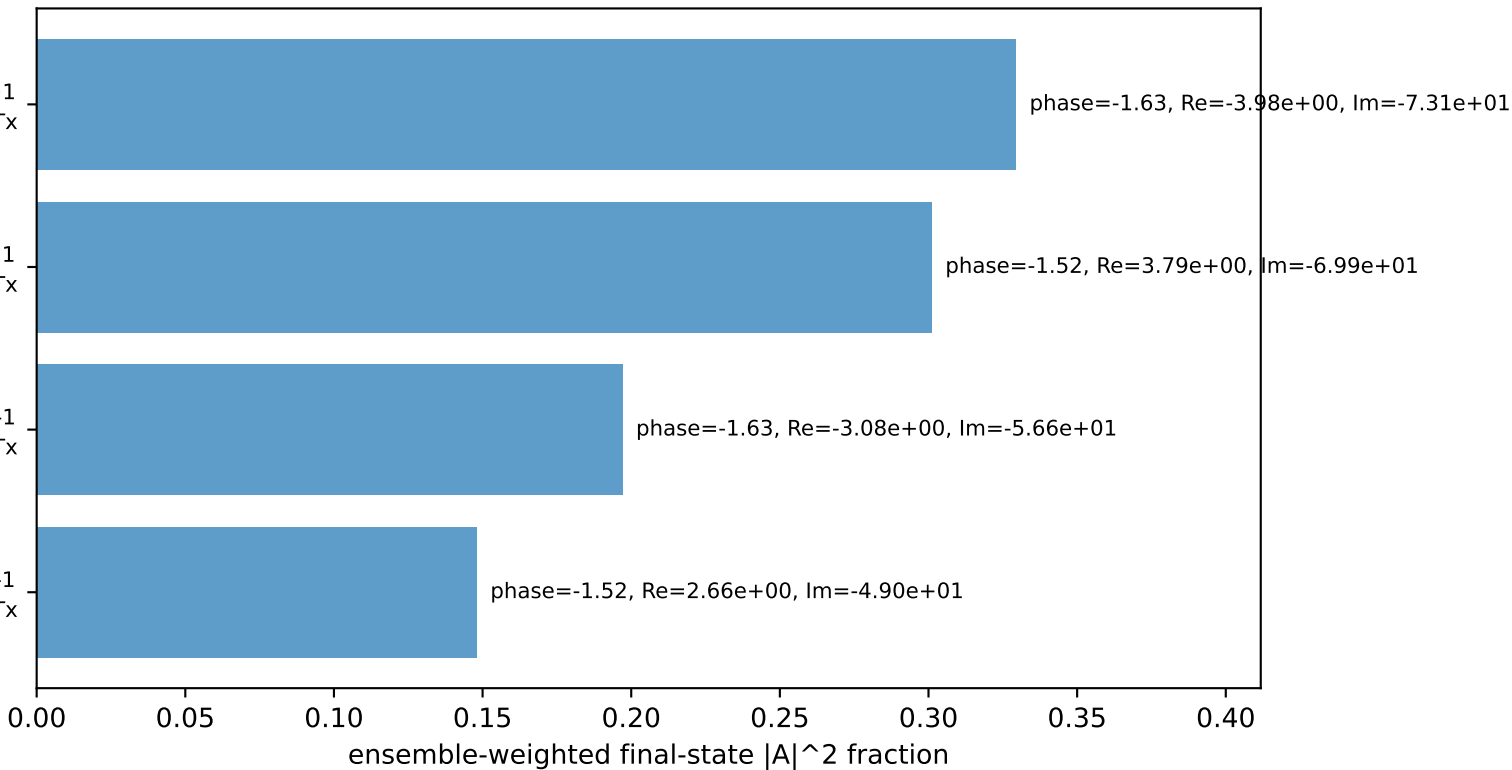
$h_e=-1, h_p=-1, h_\gamma=+1$
auxiliary lepton $h=-1$, proton Tx

$h_e=+1, h_p=-1, h_\gamma=+1$
auxiliary lepton $h=+1$, proton Tx

$h_e=-1, h_p=+1, h_\gamma=-1$
auxiliary lepton $h=-1$, proton Tx

$h_e=+1, h_p=+1, h_\gamma=-1$
auxiliary lepton $h=+1$, proton Tx

Leading initial-ensemble/final-state components (N=4, retained=97.6%)

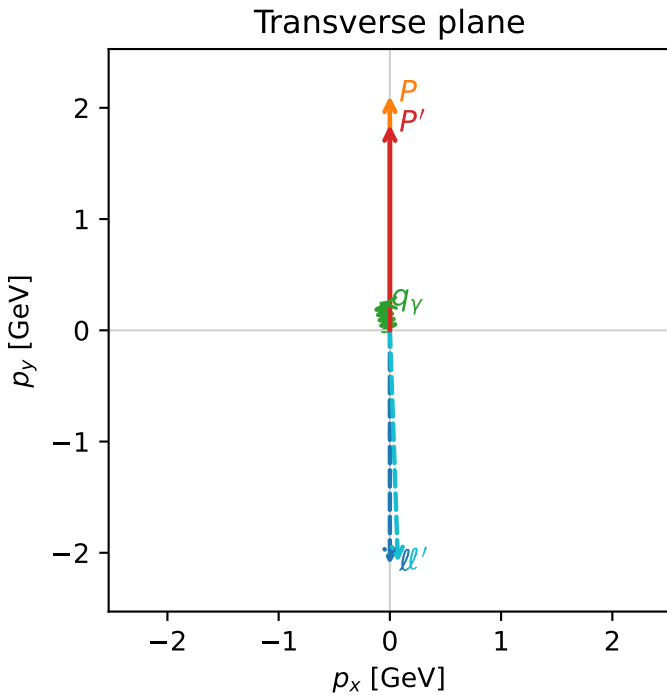
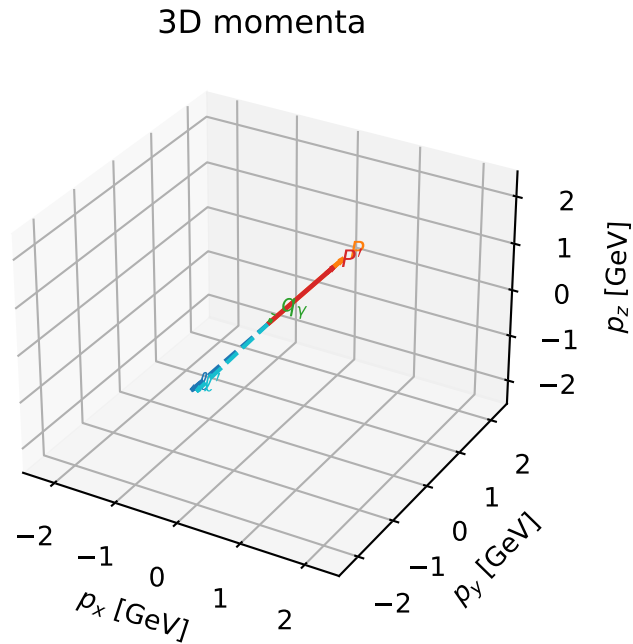


Tx proton characteristic P_W configuration

w_purity_low_egamma_tx_proton_region_1 P_W=0.325829
 region: low_Egamma_Tx_proton_region_1
 outgoing-state purity: 0.500245
 kinematic point: high_s_high_theta_in_low_Egamma
 incoming state: $\frac{1}{2} \sum_{h_\ell} \rho(h_\ell, T_{x,p})$

$s=21.5158$, $\sqrt{s}=4.63852$, $|\vec{P}|=2.10694$, $|\vec{P}'|=1.8483$
 $\theta_{in}=1.57079$, $\phi_\ell=4.71239$, $\phi_p=1.5708$
 $E_\gamma=0.25$, $\phi_\gamma=1.86532$
 $Q^2=0.00537493$, $x_B=0.0341558$, $t=-0.012321$, $W^2=1.03183$, $y=0.00801412$
 $\theta(\ell', \gamma)=165.116$ deg

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.3322$ $p_x= -0.0000$ $p_y= -2.1069$ $p_z= -0.0000$
 P $E= 2.3063$ $p_x= 0.0000$ $p_y= 2.1069$ $p_z= 0.0000$
 ℓ' $E= 2.3158$ $p_x= 0.0726$ $p_y= -2.0875$ $p_z= 0.0000$
 P' $E= 2.0727$ $p_x= 0.0000$ $p_y= 1.8483$ $p_z= 0.0000$
 q_γ $E= 0.2500$ $p_x= -0.0726$ $p_y= 0.2392$ $p_z= 0.0000$



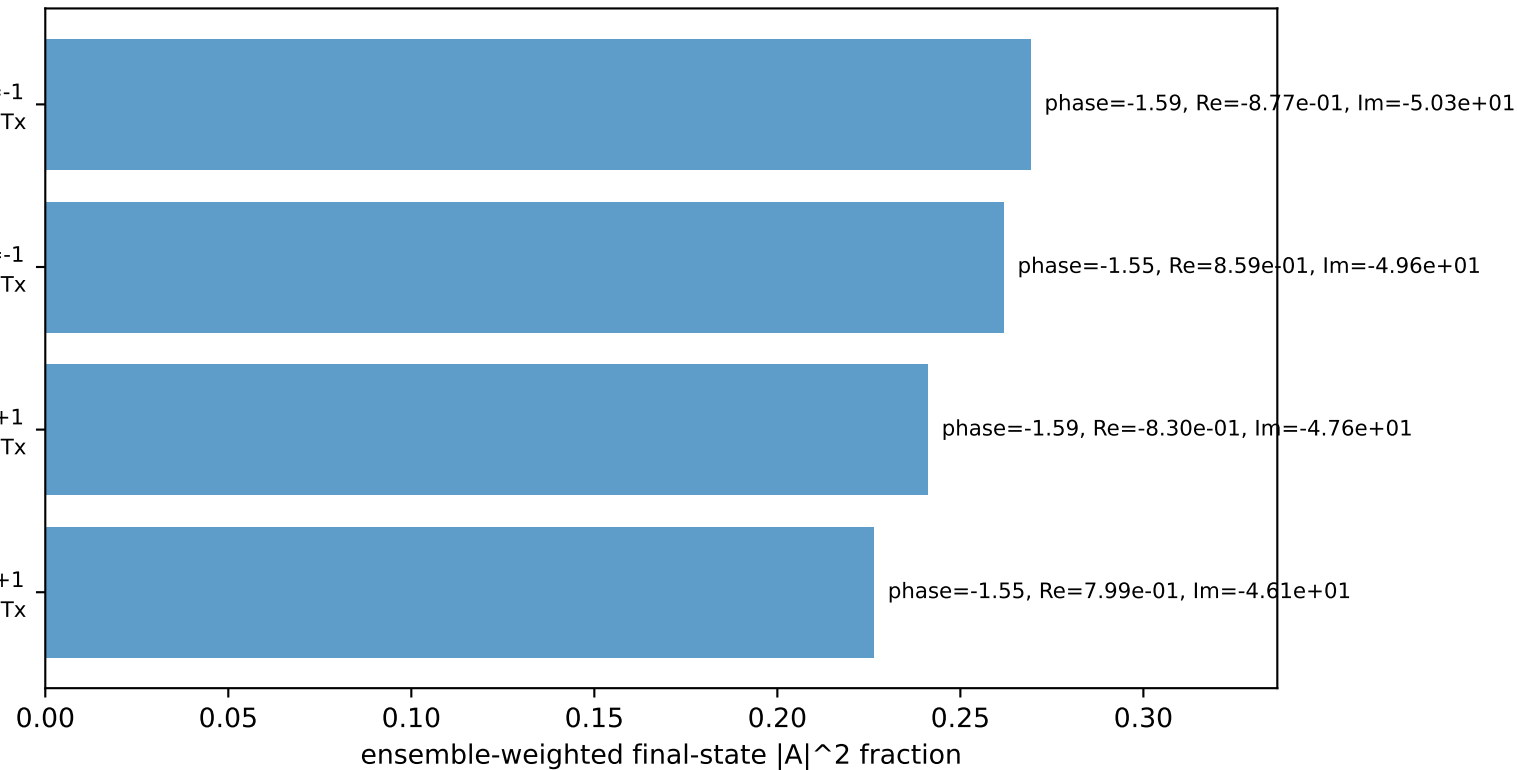
$h_e=+1, h_p=+1, h_\gamma=-1$
 auxiliary lepton $h=+1$, proton Tx

$h_e=-1, h_p=+1, h_\gamma=-1$
 auxiliary lepton $h=-1$, proton Tx

$h_e=+1, h_p=-1, h_\gamma=+1$
 auxiliary lepton $h=+1$, proton Tx

$h_e=-1, h_p=-1, h_\gamma=+1$
 auxiliary lepton $h=-1$, proton Tx

Leading initial-ensemble/final-state components (N=4, retained=99.9%)

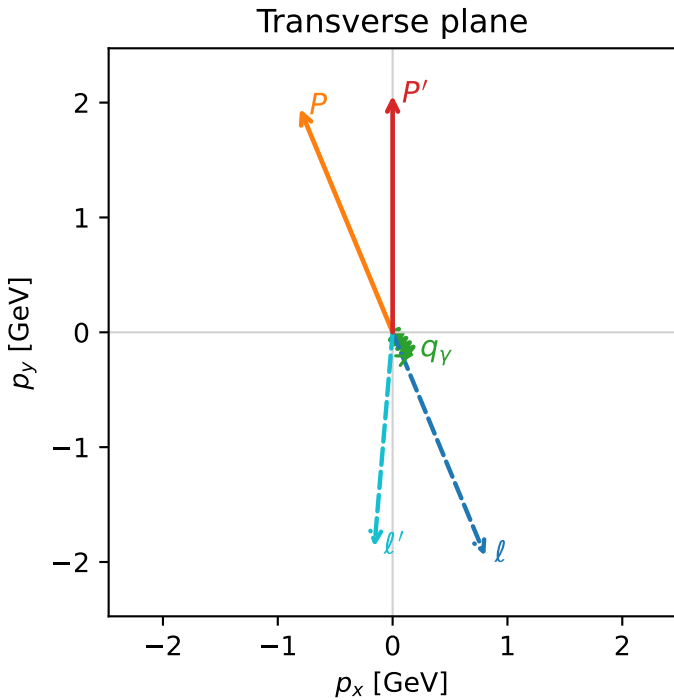
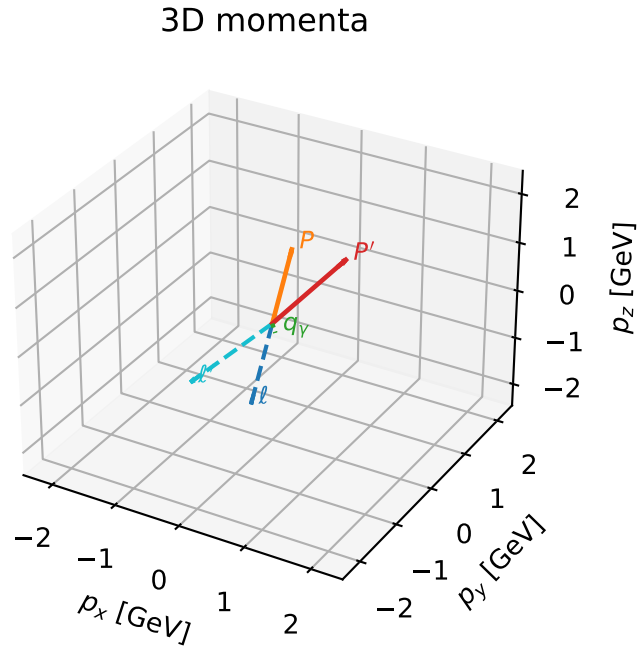


Tx proton characteristicmax P_W configuration

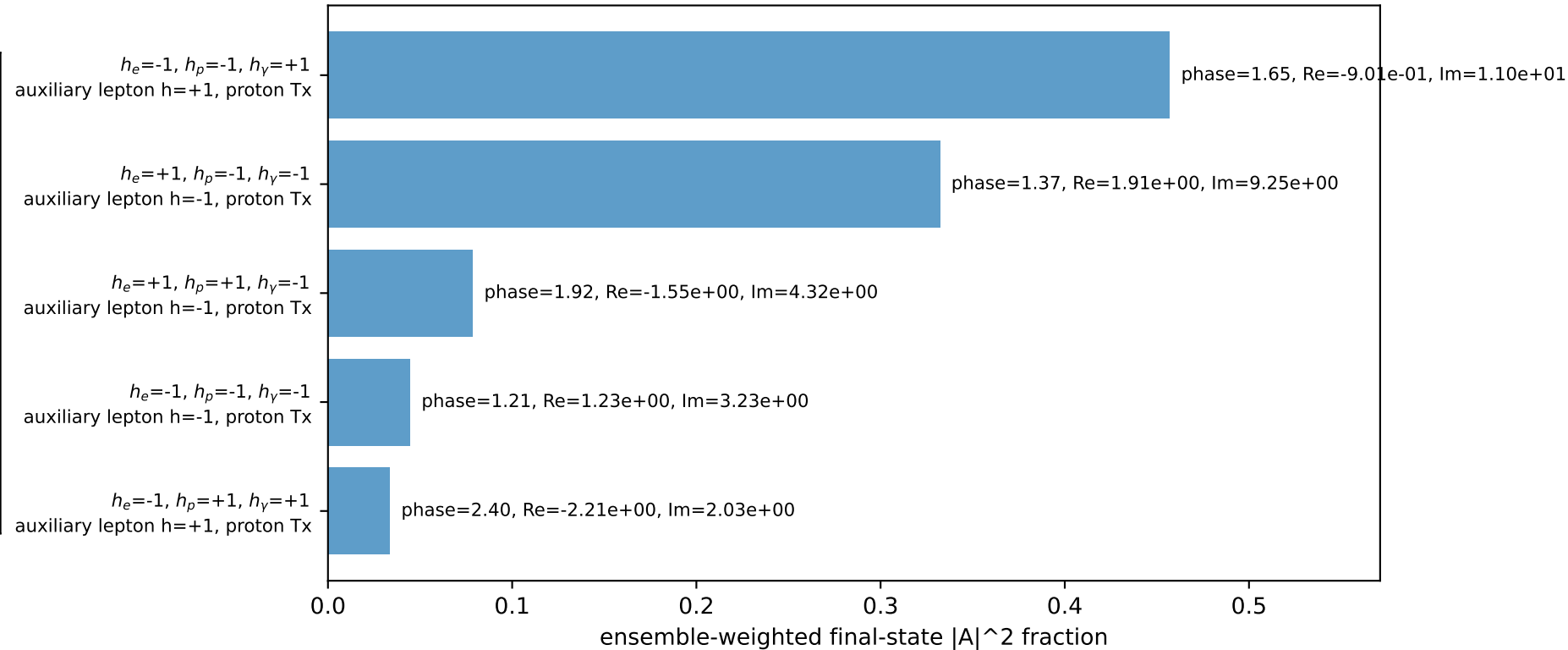
w_purity_low_egamma_tx_proton_region_2 P_W=0.291061
region: low_Egamma_Tx_proton_region_2
outgoing-state purity: 0.502596
kinematic point: high_s_high_theta_in_low_Egamma
incoming state: $\frac{1}{2}\sum_{h_\ell}\rho(h_\ell, T_{x,p})$

$s=21.5158$, $\sqrt{s}=4.63852$, $|\vec{P}|=2.10694$, $|\vec{P}'|=2.06076$
 $\theta_{in}=1.57079$, $\phi_\ell=5.10509$, $\phi_p=1.9635$
 $E_\gamma=0.25$, $\phi_\gamma=5.39961$
 $Q^2=0.894045$, $x_B=0.316741$, $t=-0.661375$, $W^2=2.80844$, $y=0.143748$
 $\theta(\ell', \gamma)=44.2292$ deg

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.3322$ $p_x= 0.8063$ $p_y= -1.9466$ $p_z= -0.0000$
 P $E= 2.3063$ $p_x= -0.8063$ $p_y= 1.9466$ $p_z= 0.0000$
 ℓ' $E= 2.1243$ $p_x= -0.1586$ $p_y= -1.8675$ $p_z= 0.0000$
 P' $E= 2.2642$ $p_x= 0.0000$ $p_y= 2.0608$ $p_z= 0.0000$
 q_γ $E= 0.2500$ $p_x= 0.1586$ $p_y= -0.1933$ $p_z= 0.0000$



Leading initial-ensemble/final-state components (N=5, retained=94.6%)



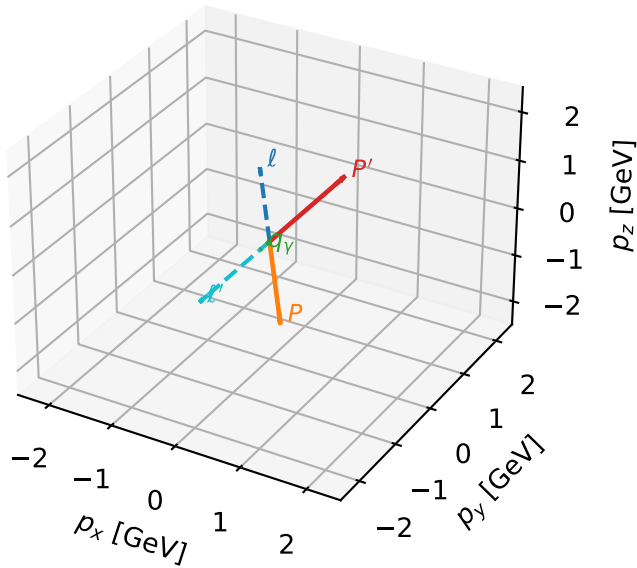
Tx proton characteristicmax P_W configuration

w_purity_low_egamma_tx_proton_region_3 P_W=0.273308
 region: low_Egamma_Tx_proton_region_3
 outgoing-state purity: 0.506177
 kinematic point: high_s_high_theta_in_low_Egamma
 incoming state: $\frac{1}{2}\sum_{h_l}\rho(h_l, T_{x,p})$

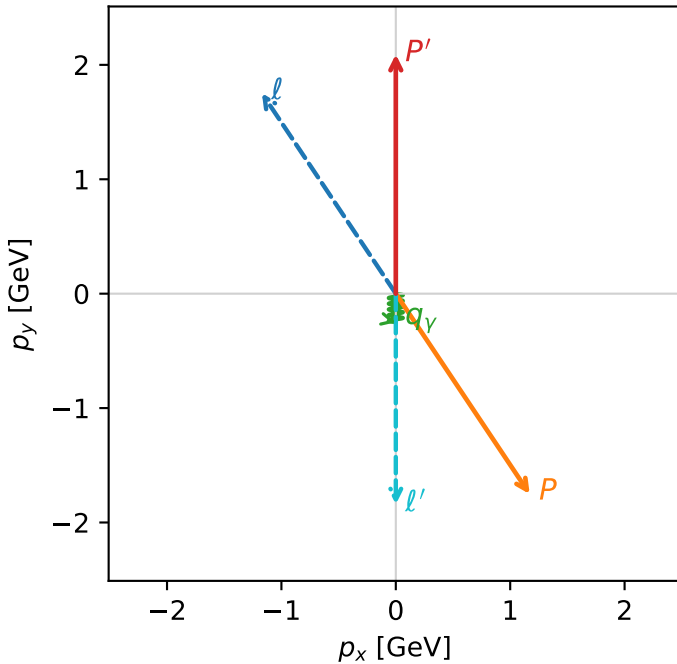
$s=21.5158$, $\sqrt{s}=4.63852$, $|\vec{P}|=2.10694$, $|\vec{P}'|=2.09195$
 $\theta_{in}=1.57079$, $\phi_l=2.15984$, $\phi_p=5.30144$
 $E_\gamma=0.25$, $\phi_\gamma=4.71239$
 $Q^2=14.2298$, $x_B=0.866504$, $t=-16.1449$, $W^2=3.07213$, $y=0.836327$
 $\theta(l', \gamma)=0$ deg

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 2.3322$ $p_x= -1.1706$ $p_y= 1.7519$ $p_z= -0.0000$
 P $E= 2.3063$ $p_x= 1.1706$ $p_y= -1.7519$ $p_z= 0.0000$
 l' $E= 2.0959$ $p_x= 0.0000$ $p_y= -1.8420$ $p_z= 0.0000$
 P' $E= 2.2926$ $p_x= 0.0000$ $p_y= 2.0920$ $p_z= 0.0000$
 q_γ $E= 0.2500$ $p_x= -0.0000$ $p_y= -0.2500$ $p_z= 0.0000$

3D momenta



Transverse plane



Leading initial-ensemble/final-state components (N=6, retained=99.2%)

